

IGCSE Chemistry CIE

YOUR NOTES



6. Chemical Reactions

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6.1 Chemical Change & Rate of Reaction

6.1.1 Physical & Chemical Changes

Physical & Chemical Changes

Physical change

- Physical changes (such as **melting** or **evaporating**) do not produce any new chemical substances
- These changes are often **easy to reverse**
- Making a mixture from 2 or more substances or dissolving a solute in a solvent are examples of physical changes as no new substances are produced and are usually relatively easy to separate

Chemical change

- During chemical changes (usually referred to as chemical reactions), **new chemical** substances are formed that have very different **properties** to the reactants
- There may be signs that a new substance has formed, such as:
 - A colour change
 - A precipitate being formed
 - Bubbles of gas being produced
- Most chemical reactions are very difficult to **reverse**
- Energy changes also accompany chemical changes and energy can be given out (**exothermic**) or taken in (**endothermic**)
 - The majority of chemical reactions are exothermic with only a small number being endothermic

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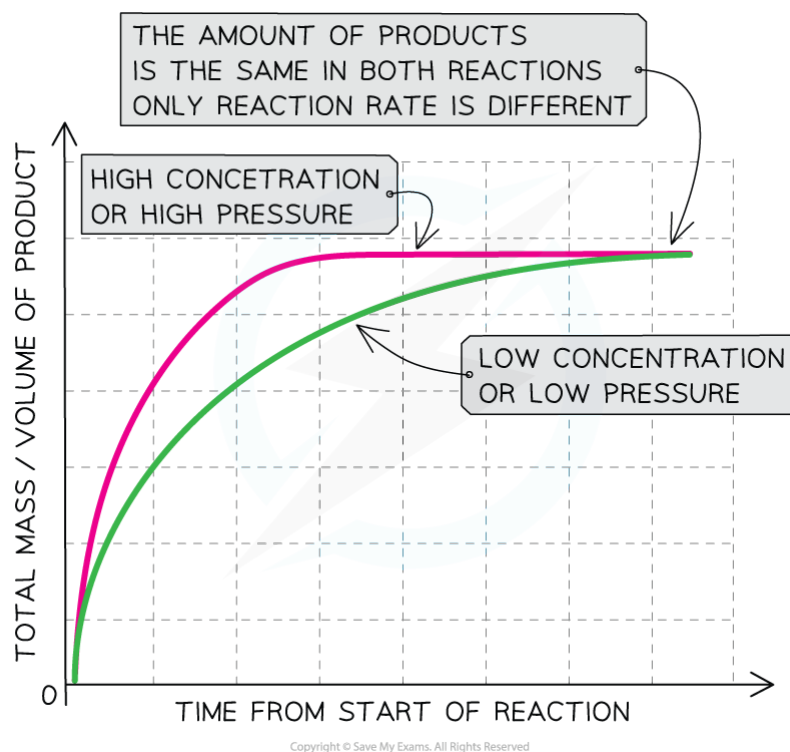
6.1.2 Rates of Reaction Factors

Rates of Reaction Factors

There are several factors that can affect the rate of a reaction. These are:

- **Concentration** of the reactants in solution or the **pressure** of reacting gases
- **Temperature** at which the reaction is carried out
- **Surface area** of solid reactants
- The use of a **catalyst**
- Changes in these factors directly influence the rate of a reaction
- It is of **economic interest** to have a higher rate of reaction as this implies a higher rate of production and hence a more efficient and sustainable process

The effect of increased concentration or pressure



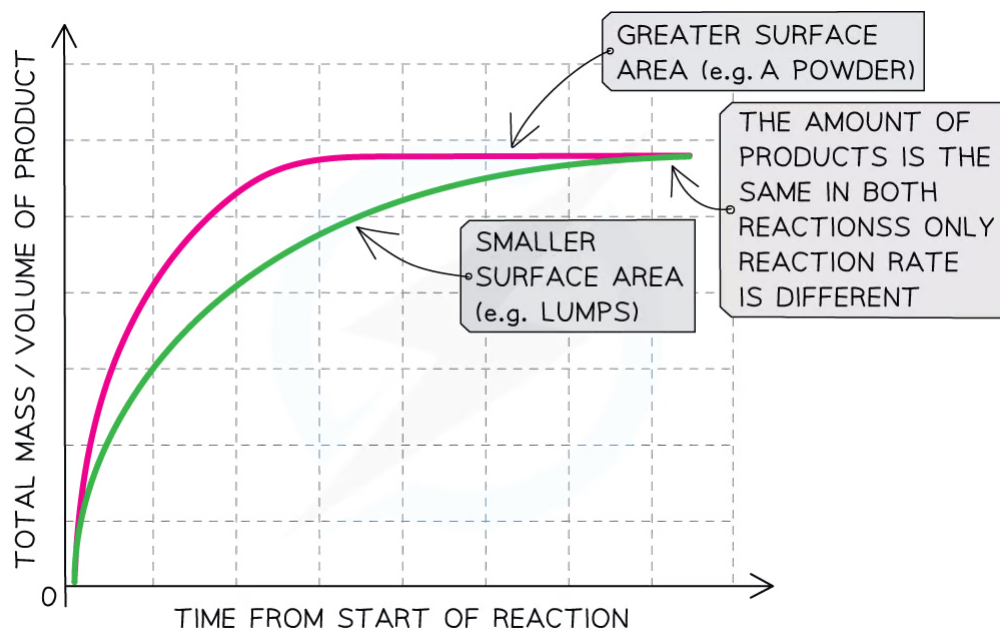
Explanation:

- Compared to a reaction with a reactant at a low concentration, the graph line for the same reaction but at a higher concentration/pressure has a steeper gradient at the start and becomes horizontal sooner
- **This shows that with increased concentration of a solution or increased pressure of a gas, the rate of reaction will increase**

The effect of surface area

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Graph showing the effect of increased surface area on the rate of reaction

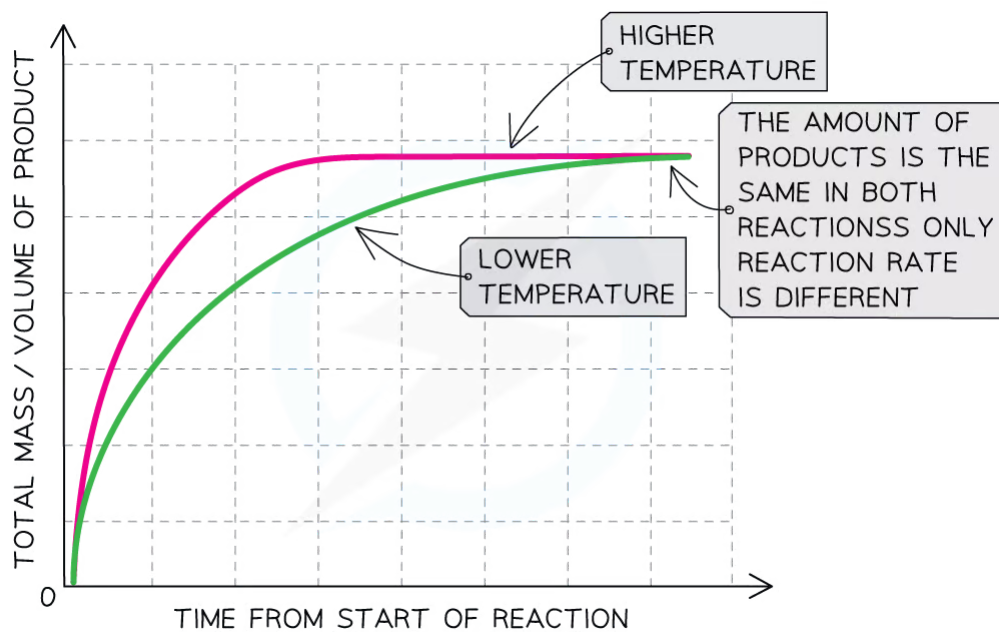
Explanation:

- Compared to a reaction with lumps of reactant, the graph line for the same reaction but with powdered reactant has a steeper gradient at the start and becomes horizontal sooner
- **This shows that with increased surface area of the solid, the rate of reaction will increase**

The effect of temperature

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Graph showing the effect of temperature on the rate of reaction

Explanation:

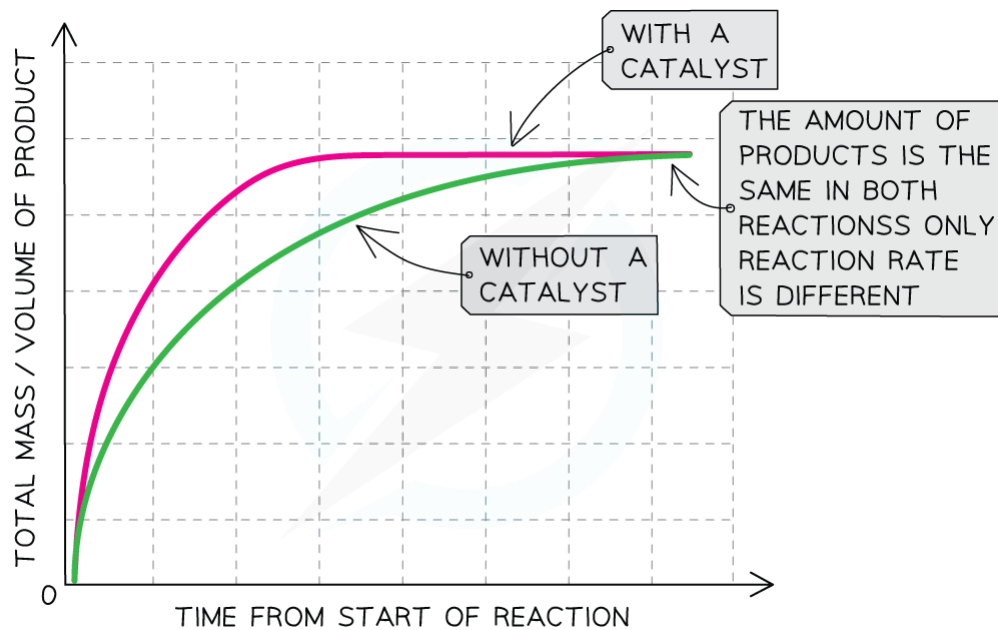
- Compared to a reaction at a low temperature, the graph line for the same reaction but at a higher temperature has a steeper gradient at the start and becomes horizontal sooner
- This shows that with increased temperature, the rate of reaction will increase**

The effect of using a catalyst

- Catalysts are substances which **speed up** the **rate** of a reaction without themselves being **altered** or **consumed** in the reaction
- The mass of a catalyst at the beginning and end of a reaction is the **same** and they do not form part of the equation

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Graph showing the effect of using a catalyst on the rate of reaction

Explanation:

- Compared to a reaction without a catalyst, the graph line for the same reaction but with a catalyst has a steeper gradient at the start and becomes horizontal sooner
- **This shows that with a catalyst, the rate of reaction will increase**

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6.1.3 Collision Theory

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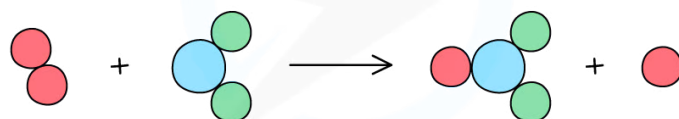


Collision Theory

- Collision theory states that in order for a reaction to occur:
 - The particles must **collide** with each other
 - The collision must have **sufficient energy** to cause a reaction i.e. enough energy to break bonds
- The minimum energy that colliding particles must have to react is known as the **activation energy**
- Collisions which result in a reaction are known as **successful collisions**
 - If they have sufficient energy (i.e. energy greater than the activation energy), they will react, and the collision will be successful
- Not all collisions result in a chemical reaction:
 - Most collisions just result in the colliding particles bouncing off each other
 - Collisions which do not result in a reaction are known as unsuccessful collisions
- Unsuccessful collisions happen when the colliding species do not have enough energy to break the necessary bonds (i.e. they collide with energy less than the activation energy)



UNSUCCESSFUL COLLISION AS PARTICLES DID NOT COLLIDE WITH ENOUGH ENERGY



SUCCESSFUL COLLISION AS PRODUCT MOLECULES WERE FORMED INDICATING THE ENERGY WITH WHICH THE PARTICLES COLLIDED WAS GREATER THAN THE ACTIVATION ENERGY

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Diagram showing a successful and an unsuccessful collision

- Increasing the number of successful collisions means that a greater proportion of reactant particles collide to form product molecules
- The number of successful collisions depends on:
 - The **number of particles per unit volume** – more particles in a given volume will produce more frequent successful collisions
 - The **frequency of collisions** – a greater number of collisions per second will give a greater number of successful collisions per second
 - The **kinetic energy of the particles** – greater kinetic energy means a greater proportion of collisions will have an energy that exceeds the activation energy

and the more frequent the collisions will be as the particles are moving quicker, therefore, more collisions will be successful

- The **activation energy** – fewer collisions will have an energy that exceeds higher activation energy and fewer collisions will be successful
- These all have an impact on the rate of reaction which is dependent on the number of successful collisions per unit of time

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6.1.4 Explaining Rates Using Collision Theory

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Explaining Rates Using Collision Theory

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Temperature

- **Increasing the temperature will increase the rate of reaction**
- This is because the particles will have more kinetic energy than the required activation energy
- Therefore there will be more frequent collisions **and** a higher proportion of particles have energy greater than the activation energy
- This causes more successful collisions per second, increasing the rate of reaction
- The effect of temperature on collisions is not so straightforward as concentration or surface area; a small increase in temperature causes a large increase in rate
- For aqueous and gaseous systems, a rough rule of thumb is that for every 10 °C increase in temperature, the rate of reaction approximately doubles

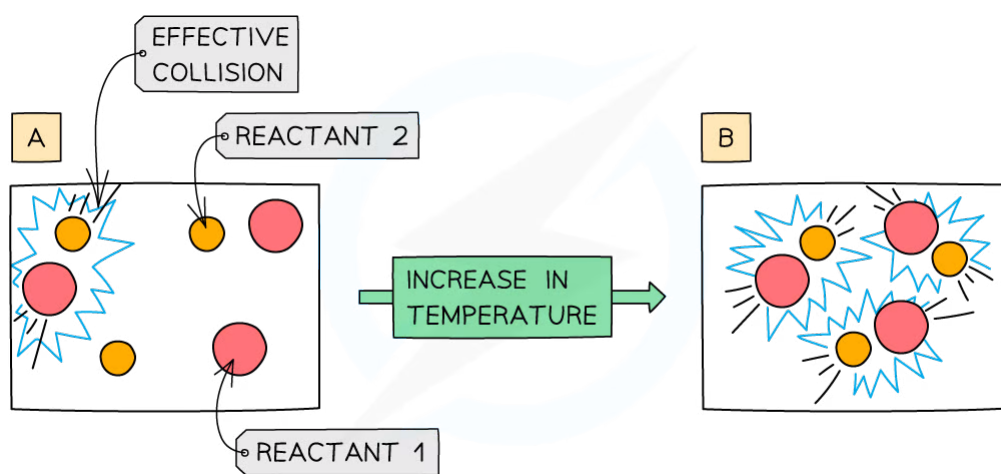


Diagram showing the increased kinetic energy that particles have at higher temperatures

Concentration

- **Increasing the concentration of a solution will increase the rate of reaction**
- This is because there will be more reactant particles in a given volume, allowing more frequent and successful collisions per second, increasing the rate of reaction
- For a gaseous reaction, increasing the pressure has the same effect as the same number of particles will occupy a smaller space, increasing the concentration
- If you double the number of particles you will double the number of collisions per second

- The number of collisions is **proportional** to the number of particles present

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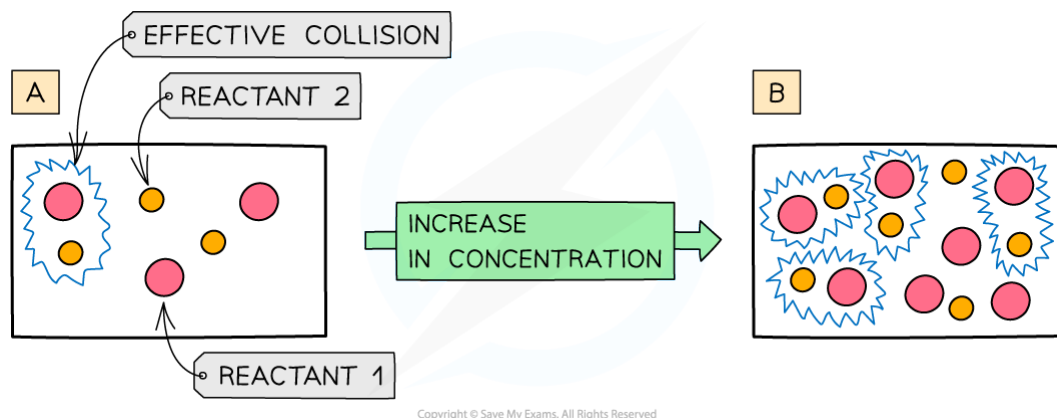
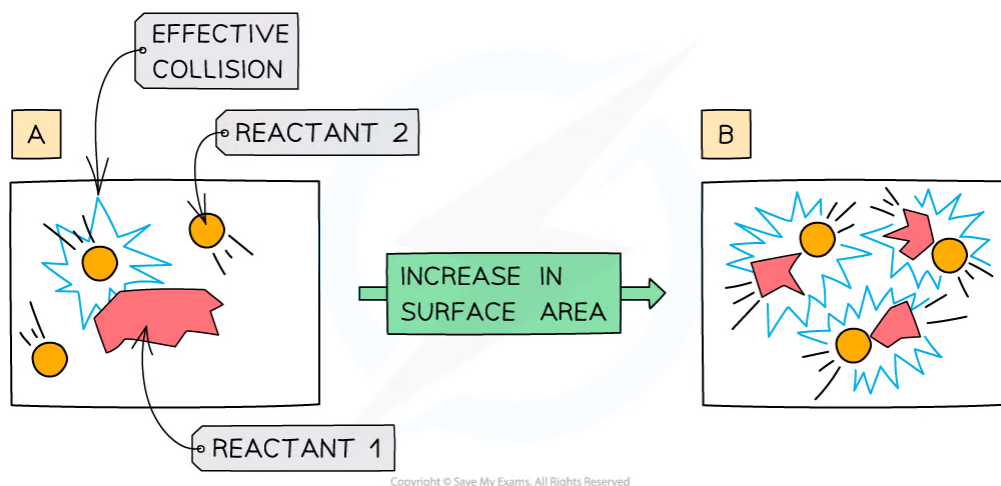


Diagram showing the decrease in space between particles at higher concentrations

Surface Area

- Increasing the surface area of a solid will increase the rate of reaction
- This is because more **surface area** of the **particles** will be exposed to the other reactant, producing a higher number of collisions per second
- If you double the surface area you will double the number of collisions per second



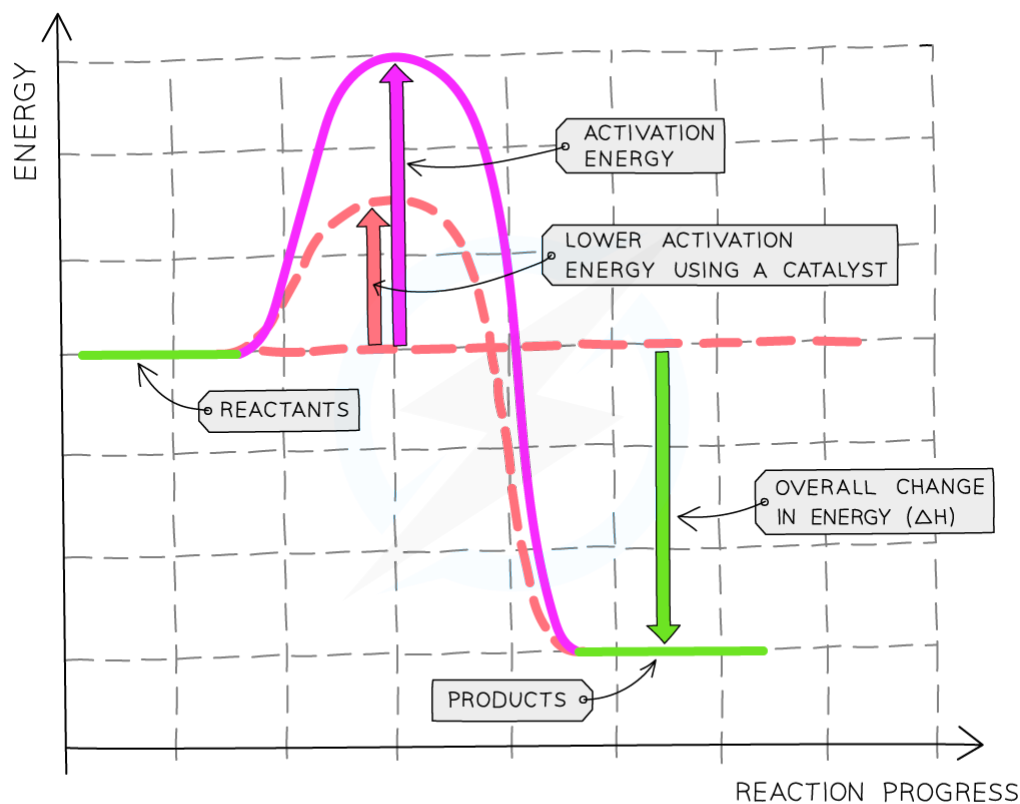
An increase in surface area means more collisions per second

Catalysts

- Catalysts are substances which **speed up** the **rate** of a reaction without themselves being **altered** or **consumed** in the reaction

- The mass of a catalyst at the beginning and end of a reaction is the **same** and they do not form part of the equation
- Different processes require different types of catalysts but they all work on the same principle of providing a **different pathway** for the reaction to occur that has a **lower activation energy**
- This means a higher proportion of the reactant particles have energy greater than the activation energy and will result in more successful collisions per second
- An important industrial example is iron, which is used to catalyse the **Haber Process** for the production of ammonia
- Iron **beads** are used to increase the **surface area** available for catalysis
- Enzymes are **biological** catalysts, they work best at specific temperature and pH ranges
- Normally only **small amounts** of catalysts are needed to have an effect on a reaction

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A catalyst lowers the activation energy of a reaction by providing an alternative reaction pathway



Exam Tip

When answering questions on the effect of concentration on the rate of reaction, you should mention that there are more particles **per unit volume** (usually cm^3) and this causes an increase in the rate of collisions.

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6.1.5 Investigating The Rate of a Reaction

Investigating The Rate of a Reaction

- To measure the **rate of a reaction**, we need to be able to measure either how quickly the reactants are used up or how quickly the products are formed
- The method used for measuring depends on the substances involved
- There are a number of ways to measure a reaction rate in the lab; they all depend on some property that changes during the course of the reaction
- That property is taken to be **proportional** to the concentration of the reactant or product, e.g., colour, mass, volume
- - faster reactions can be easier to measure when the reaction is over, by averaging a collected measurement over the course of the reaction some reaction rates can be measured as the reaction proceeds (this generates more data)
- Three commonly used techniques are:
 - **measuring mass loss on a balance**
 - **measuring the volume of a gas produced**
 - **measuring a reaction where there is a colour change at the end of the reaction**

Investigating the effect of surface area on the rate of reaction

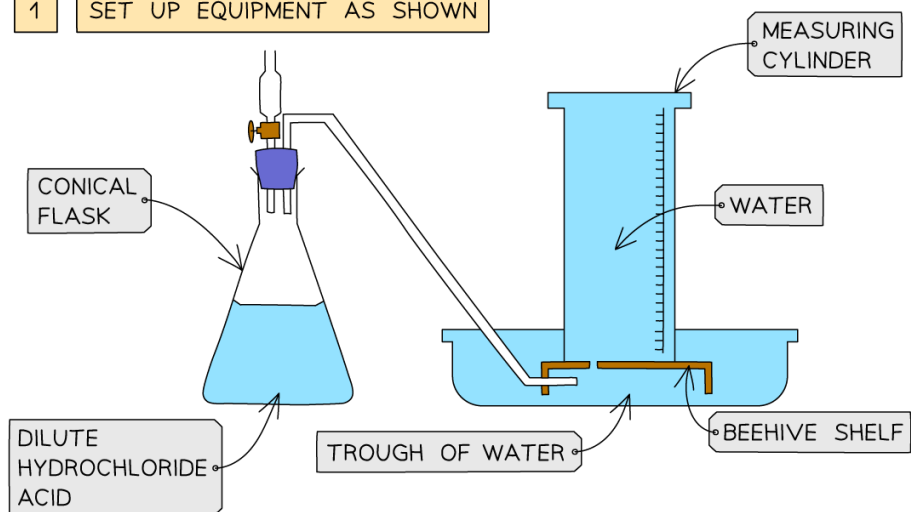
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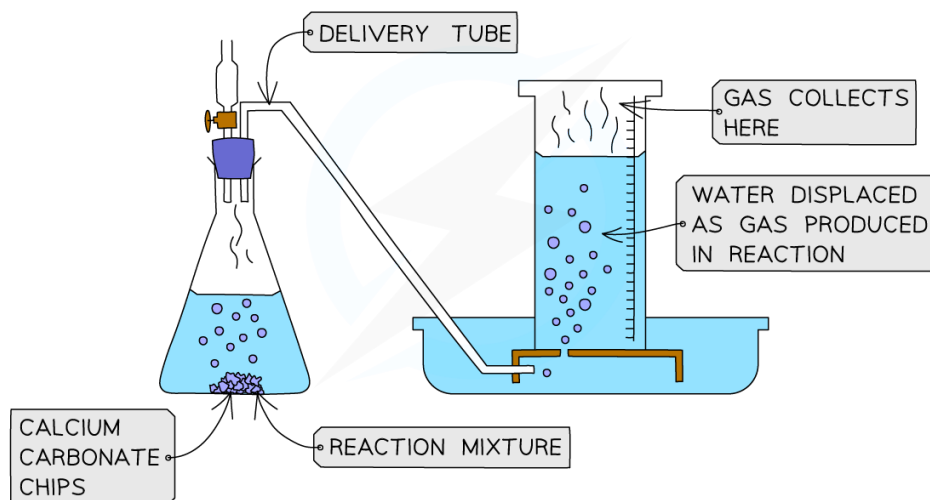
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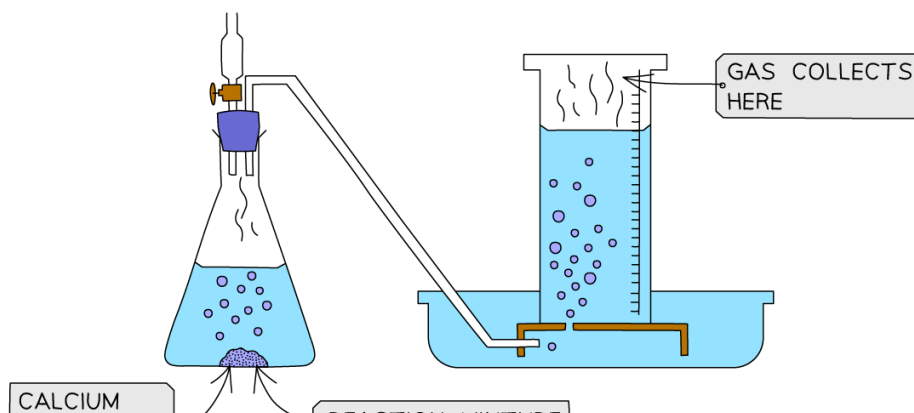
1 SET UP EQUIPMENT AS SHOWN



2 ADD CALCIUM CARBONATE CHIPS TO ACID, AFTER A FIXED TIME RECORD THE VOLUME OF WATER DISPLACED



3 REPEAT, USING DIFFERENT FORMS OF CALCIUM CARBONATE



CARBONATE
POWDER

REACTION MIXTURE

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Diagram showing the process of downwards displacement to investigate the effect of the surface area of a solid on the rate of reaction

Method:

- Add dilute hydrochloric acid into a conical flask
- Use a delivery tube to connect this flask to a measuring cylinder upside down in a bucket of water (downwards displacement)
- Add calcium carbonate chips into the conical flask and quickly put the bung back into the flask
- Measure the volume of gas produced in a fixed time using the measuring cylinder
- Repeat with different sizes of calcium carbonate chips (lumps, crushed and powdered)

Result:

- Smaller sizes of chips cause an increase in the surface area of the solid, so the rate of reaction will increase
- This is because more surface area of the particles will be exposed to the other reactant so there will be more frequent and successful collisions, increasing the rate of reaction

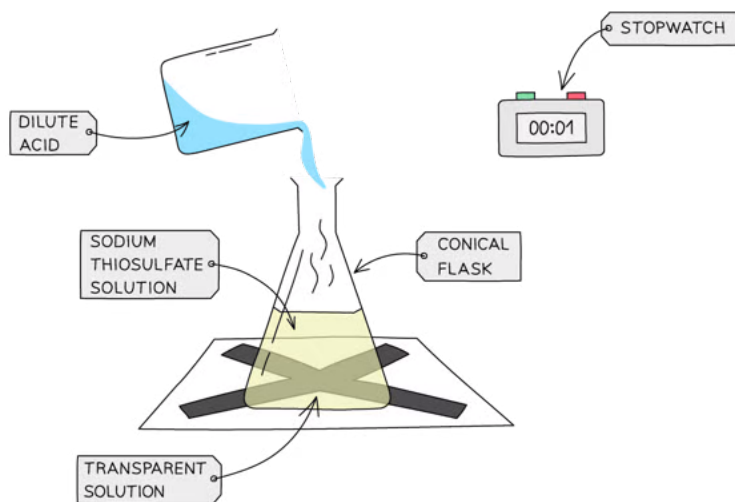
Effect of concentration of a solution on the rate of reaction:

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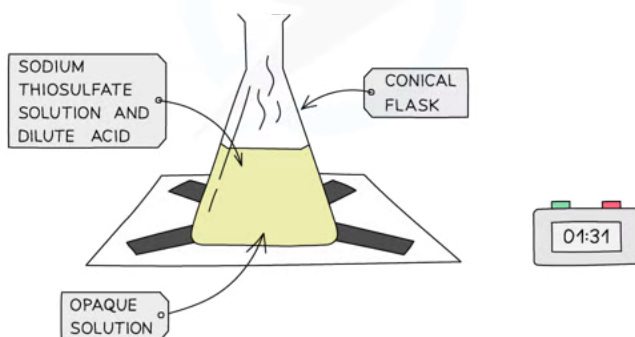
1

ADD ACID TO SODIUM THIOSULFATE SOLUTION, START TIMER.



2

STOP TIMER WHEN 'X' IS NO LONGER VISIBLE. RECORD TIME.



3

REPEAT STEPS 1-2 WITH DIFFERENT CONCENTRATIONS OF SODIUM THIOSULFATE (MADE BY MIXING DIFFERENT VOLUMES OF WATER AND SODIUM THIOSULFATE)

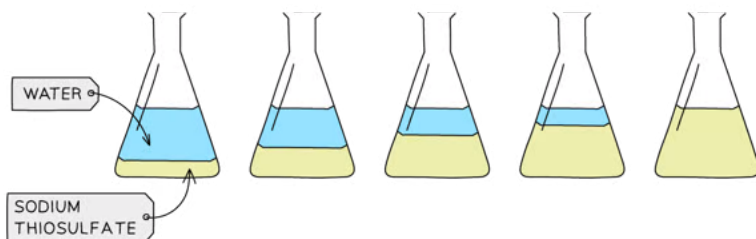


Diagram showing the apparatus needed to investigate the effect of concentration on the rate of reaction

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**Method:**

- Measure 50 cm³ of sodium thiosulfate solution into a flask
- Measure 5 cm³ of dilute hydrochloric acid into a measuring cylinder
- Draw a cross on a piece of paper and put it underneath the flask
- Add the acid into the flask and immediately start the stopwatch
- Look down at the cross from above and stop the stopwatch when the cross can no longer be seen
- Repeat using different concentrations of sodium thiosulfate solution (mix different volumes of sodium thiosulfate solution with water to dilute it)

Result:

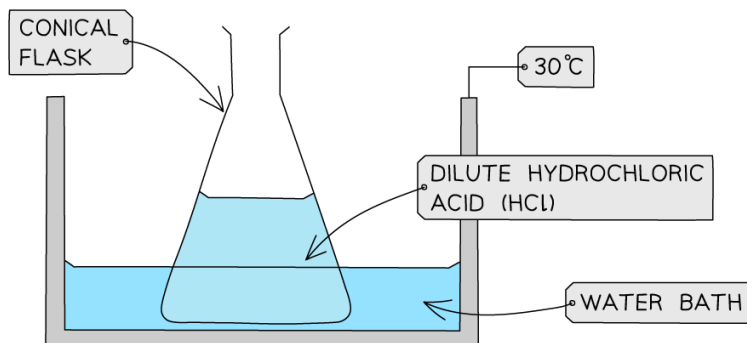
- With an increase in the concentration of a solution, the rate of reaction will increase
- This is because there will be more reactant particles in a given volume, allowing more frequent and successful collisions, increasing the rate of reaction

Effect of temperature on the rate of reaction:

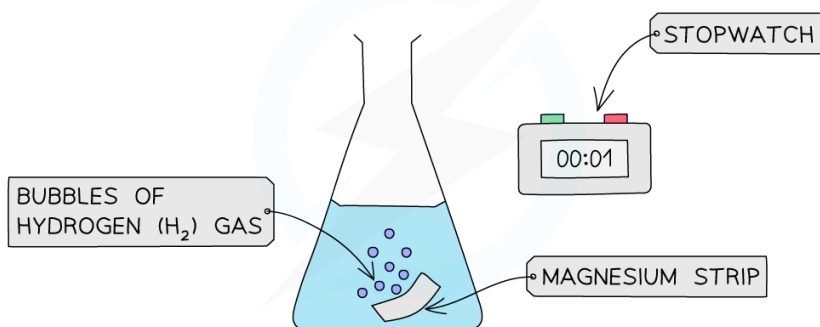
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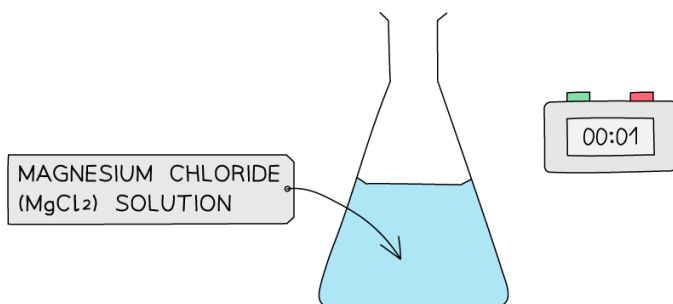
1 HEAT DILUTE HYDROCHLORIC ACID TO A DESIRED TEMPERATURE



2 ADD A STRIP OF MAGNESIUM TO THE DILUTE HCl AND START A STOPWATCH



3 WHEN THE MAGNESIUM STRIP HAS DISSOLVED, STOP THE STOPWATCH AND RECORD THE TIME



4 REPEAT EXPERIMENT AT DIFFERENT TEMPERATURES

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Diagram showing the apparatus needed to investigate the effect of temperature on the rate of reaction

Method:

- Dilute hydrochloric acid is heated to a set temperature using a water bath

- Add the dilute hydrochloric acid into a conical flask
- Add a strip of magnesium and start the stopwatch
- Stop the time when the magnesium fully reacts and disappears
- Repeat at different temperatures and compare results

Result:

- With an increase in the temperature, the rate of reaction will increase
- This is because the particles will have more kinetic energy than the required activation energy, therefore more frequent and successful collisions will occur, increasing the rate of reaction

Effect of a catalyst on the rate of reaction:

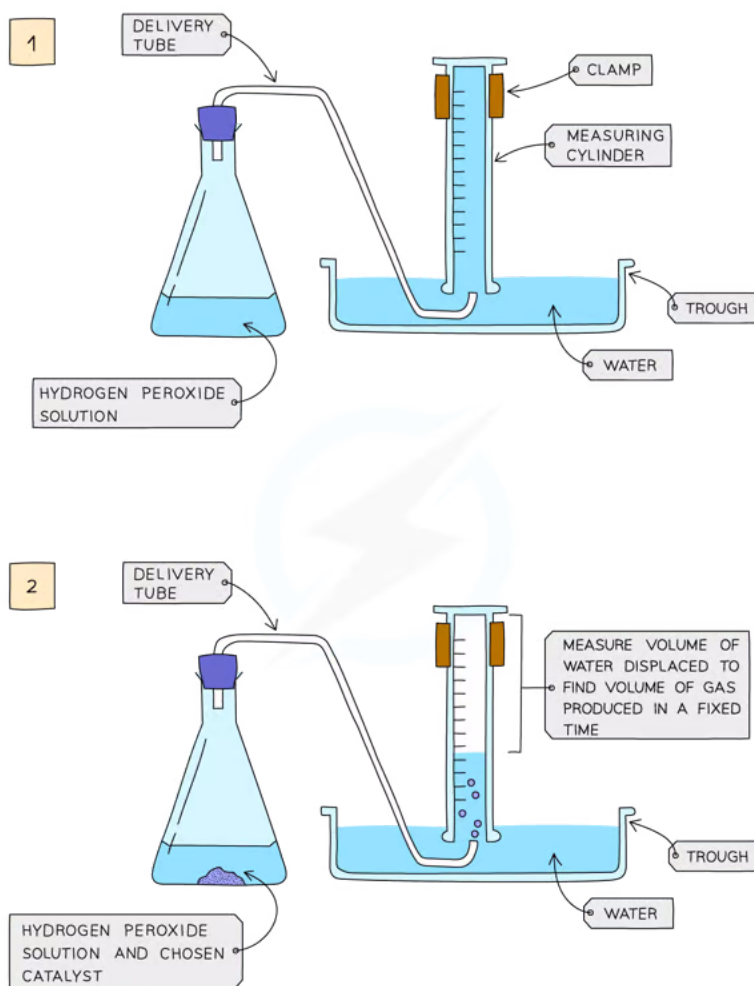


Diagram showing the apparatus needed to investigate the effect of a catalyst on the rate of reaction

Method:

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- Add hydrogen peroxide into a conical flask
- Use a delivery tube to connect this flask to a measuring cylinder upside down in a tub of water (downwards displacement)
- Add the catalyst manganese(IV) oxide into the conical flask and quickly place the bung into the flask
- Measure the volume of gas produced in a fixed time using the measuring cylinder
- Repeat experiment without the catalyst of manganese(IV) oxide and compare results

Result:

- Using a catalyst will increase the rate of reaction
- The catalyst will provide an alternative pathway requiring lower activation energy so more colliding particles will have the necessary activation energy to react
- This will allow more frequent and successful collisions, increasing the rate of reaction

Monitoring changes in mass

- Many reactions involve the production of a gas which will be released during the reaction
- The gas can be collected and the volume of gas monitored as per some methods above
- Alternatively, the reaction can be performed in an **open flask** on a balance to measure the loss in mass of reactant
- Cotton wool is usually placed in the mouth of the flask which allows gas out but prevents any materials from being ejected from the flask (if the reaction is vigorous)

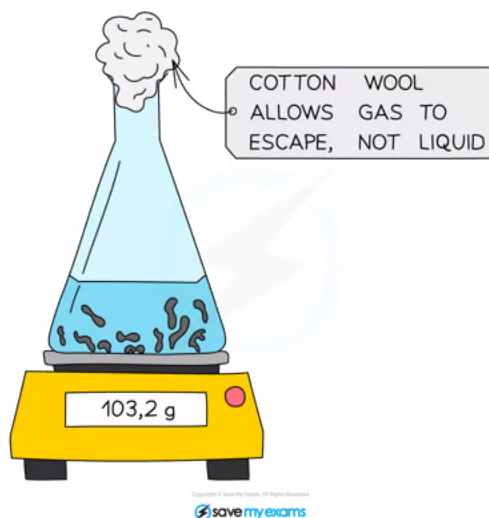


Diagram showing the set-up for measuring the rate of reaction by loss in mass

- This method is not suitable for hydrogen and other gases with a small relative formula mass, M_r , as the loss in mass may be too small to measure



Exam Tip

There are many different methods of investigating the rate of reaction. Another method of gas collection you may see uses a gas syringe.

Those students taking the extended course may be required to devise and evaluate methods of investigating rates of reaction.

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Evaluating Investigations of Rates of Reactions

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- When investigating rates of reaction, there are a number of different methods that can be used to carry out the same investigation
- Evaluating what is the best method to use is part of good experimental planning and design
- This means appreciating some of the advantages and disadvantages of the methods available

Table showing some Examples of Advantages and Disadvantages of Methods of Investigating Rates of Reaction

Method Choice	Advantage	Disadvantage
Disappearing cross experiment	◦ Simple experiment with no specialist equipment required	◦ Difficult to determine when cross is obscured ◦ Different people will determine the cross to have disappeared at different levels of cloudiness
Collecting gas using a gas syringe	◦ All the gas collected is from the reaction ◦ Gas syringes are easy to set up and keep gas dry	◦ Gas syringes are expensive and fragile ◦ The pistons can stick ◦ Limited volume can be collected ◦ Gas can be lost if the bung is not put onto the top of the flask quickly
Collecting gas using an inverted measuring cylinder	◦ The experiment uses common lab equipment	◦ The delivery tube beneath the measuring cylinder can come away ◦ It can be difficult to read the scale especially when the reaction is vigorous with a large number of bubbles produced rapidly ◦ Gas can be lost if the bung is not put onto the top of the flask quickly
Measuring the loss of mass of reactants	◦ Simple experiment to carry out with no specialist equipment	◦ Not suitable for gases with a small Mr

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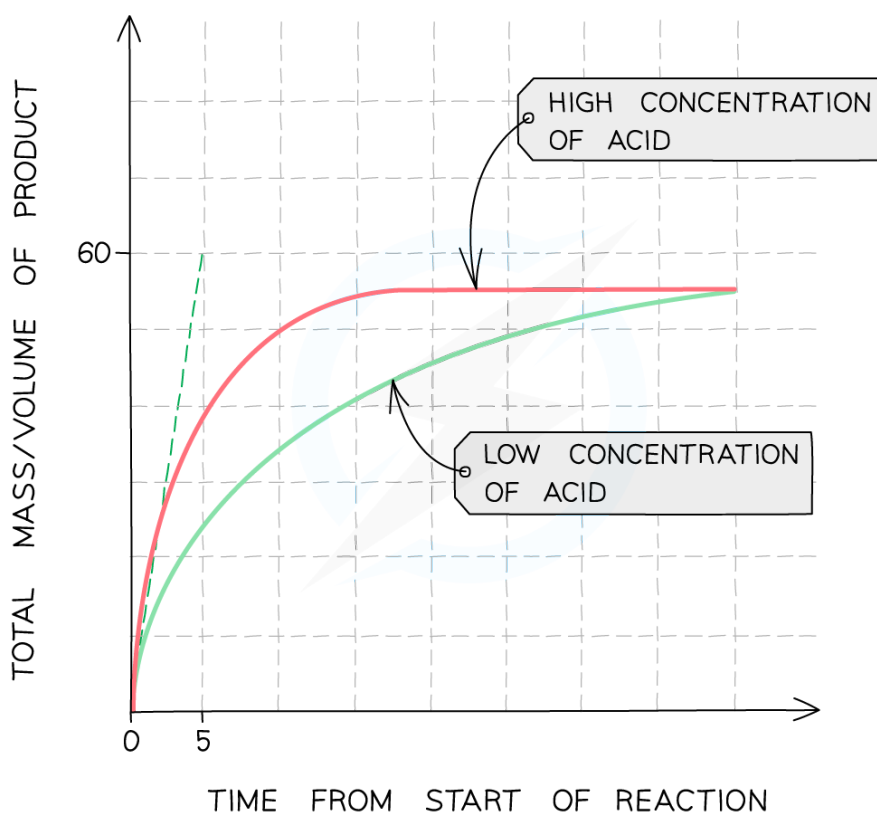
6.1.6 Interpreting Data

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Interpreting Data

- Data recorded in rate studies is used to **plot graphs** to calculate the rate of a reaction
- Plotting a graph until the completion of the reaction shows how the rate changes with time
- Over time the rate of reaction slows as the reactants are being used up so the line becomes less steep and eventually becomes **horizontal**, indicating the reaction has finished
- You can plot more than one run of a variable on the same graph making it easier to see how the variable influences the rate
 - For example, plotting the effect of concentration on a reaction between the acid and marble chips



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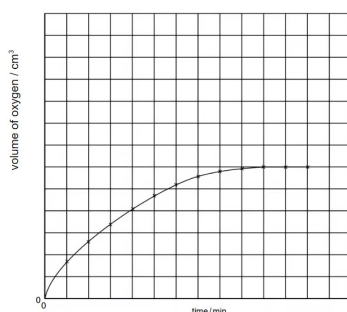
- The steeper the curve, the faster the rate of the reaction
- The curve is steepest initially so the rate is quickest at the beginning of the reaction
- As the reaction progresses, the concentration of the reactants decreases and the rate decreases shown by the curve becoming less steep



- When one of the reactants is used up, the reaction stops, the rate becomes zero and the curve levels off to a horizontal line
- The amount of product formed in a reaction is determined by the limiting reactant:
 - If the amount of limiting reactant increases, the amount of product formed increases
 - If the amount of the reactant in excess increases, the amount of product remains the same
- Drawing a **tangent** to the slope allows you to show the gradient at any point on the curve
- The steeper the slope, the quicker the rate of reaction
- The volume of a gaseous product would increase to a maximum over time, so the line levels out indicating the reaction is over
- Since the volume and mass would be proportional, this could also be a graph of the mass of product versus time

? Worked Example

0.2 g of manganese(IV) oxide was added to 25 cm³ of 0.1 mol/dm³ hydrogen peroxide solution. The volume of oxygen produced every minute was recorded and the results are shown on the graph.



The experiment was repeated using the same mass of manganese(IV) oxide and at the same temperature but using 25 cm³ of 0.2 mol/dm³ of hydrogen peroxide solution.

Sketch the curve for the results of this experiment on the same grid.

Answer

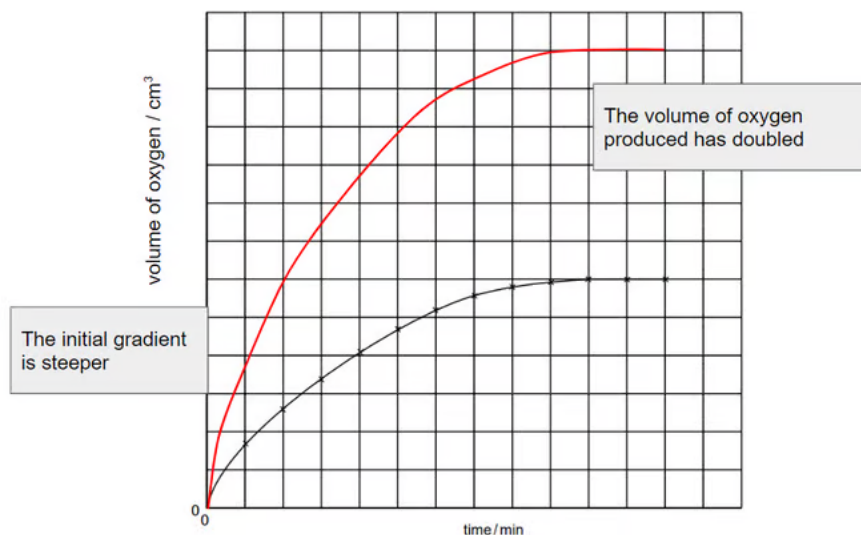
Step 1 – Deduce how the initial gradient will be different from the original graph

The hydrogen peroxide solution is twice as concentrated so the rate of reaction will be greater and the initial gradient will be steeper

Step 2 – Deduce how much product will be formed compared to the original experiment

The amount of hydrogen peroxide determines the amount of oxygen produced. In the 2nd experiment, there are twice as many hydrogen peroxide molecules in the same volume so the amount of oxygen gas produced will be doubled

Step 3 – Sketch the graph



Exam Tip

If the amount of reactant used up is being monitored, then the graph will fall with the steepest gradient at the start, becoming less steep until it levels off to a horizontal line.

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6.2 Reversible Reactions & Equilibrium

6.2.1 Reversible Reactions

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Reversible Reactions

Reversible reactions

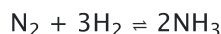
- Some reactions go to **completion**, where the reactants are used up to form the product molecules and the reaction stops when all of the reactants are used up
- In reversible reactions, the product molecules can themselves react with each other or decompose and form the reactant molecules again
- It is said that the reaction can occur in **both directions**: the forward reaction (which forms the products) and the reverse reaction (which forms the reactants)

Chemical equations for reversible reactions

- When writing chemical equations for reversible reactions, two arrows are used to indicate the forward and reverse reactions
- Each one is drawn with just half an arrowhead – the top one points to the right, and the bottom one points to the left: $\rightleftharpoons$

Example

- The reaction for the **Haber process** which is the production of ammonia from hydrogen and nitrogen:

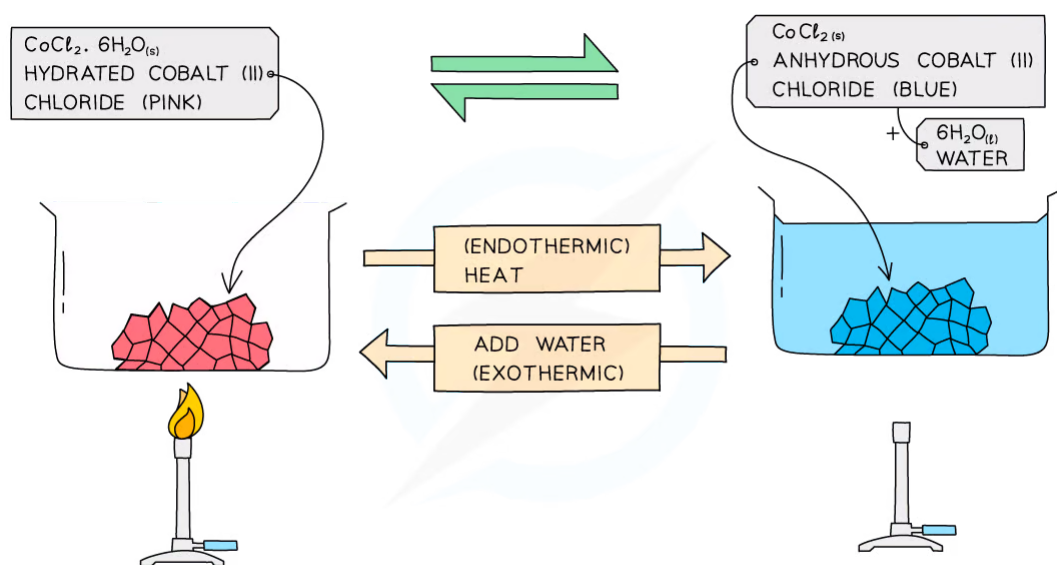


Hydrated and anhydrous salts

- Hydrated salts are salts that contain **water of crystallisation** which affects their molecular shape and colour
- Water of crystallisation is the water that is stoichiometrically included in the structure of some salts during the crystallisation process
- A common example is copper(II) sulfate which crystallises forming the salt copper(II) sulfate pentahydrate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- Water of crystallisation is indicated with a dot written in between the salt molecule and the surrounding water molecules
- Anhydrous salts are those that have lost their water of crystallisation, usually by heating, in which the salt becomes **dehydrated**

Dehydration of hydrated cobalt(II) chloride:

hydrated cobalt(II) chloride $\rightleftharpoons$ anhydrous cobalt(II) chloride + water



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Diagram showing the dehydration of hydrated cobalt(II) chloride

Hydration of cobalt(II) chloride

- When anhydrous **blue** cobalt(II) chloride crystals are added to water they turn **pink** and the reaction is **reversible**
- When the cobalt(II) chloride crystals are heated in a test tube, the **pink** crystals turn back to the **blue** colour again as the water of crystallisation is lost
- The form of cobalt(II) chloride in the crystals that are pink is known as hydrated cobalt(II) chloride because it contains water of crystallisation
- When hydrated cobalt(II) chloride is heated, it loses its water of crystallisation and turns into anhydrous cobalt(II) chloride:



Exam Tip

The hydration of CoCl_2 (and CuSO_4) are chemical tests which are commonly used to detect the presence of water. You should remember the equations and colour changes:

- $\text{CoCl}_2 + 6\text{H}_2\text{O} \rightleftharpoons \text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ **Blue to pink**
- $\text{CuSO}_4 + 5\text{H}_2\text{O} \rightleftharpoons \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ **White to blue**

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6.2.2 Equilibrium

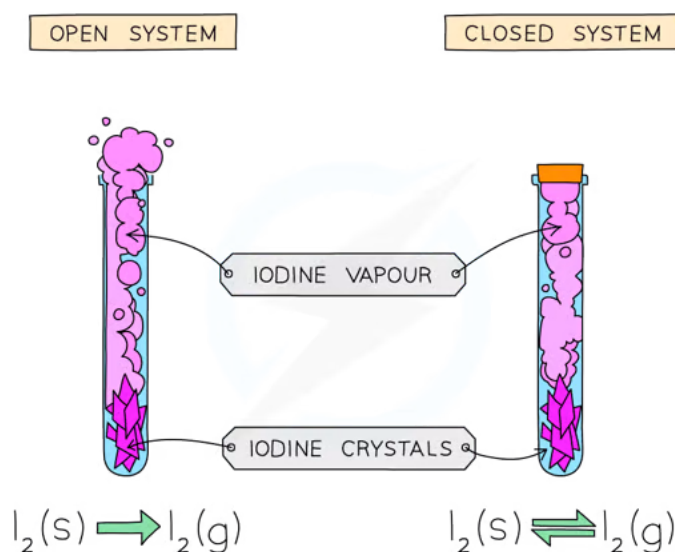
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The Concept of Equilibrium

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- We have already seen that a reversible reaction is one that occurs in **both** directions
- When during the course of reaction, the **rate of the forward reaction equals the rate of the reverse reaction**, then the overall reaction is said to be in a state of **equilibrium**
- Equilibrium is **dynamic** e.g. the molecules on the left and right of the equation are **changing** into each other by chemical reactions constantly and at the same rate
- The concentration of reactants and products remains **constant** (given there is no other change to the system such as temperature and pressure)
- It only occurs in a **closed system** so that none of the participating chemical species are able to leave the reaction vessel

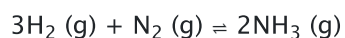


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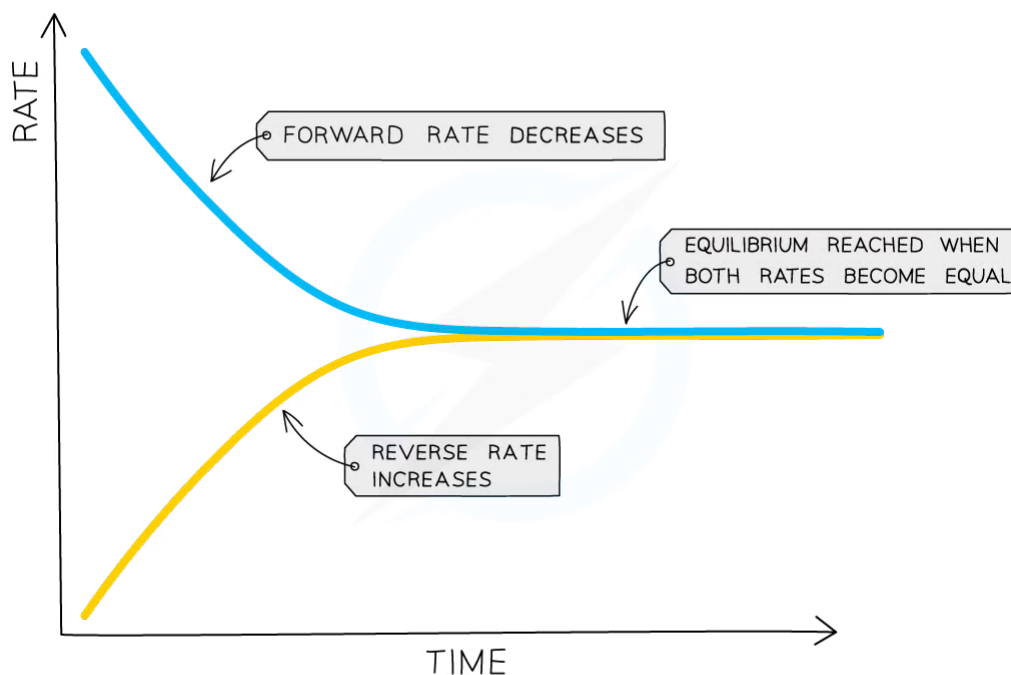
Equilibrium can only be reached in a closed vessel which prevents reactants or products from escaping the system

- An example of dynamic equilibrium is the reaction between H_2 and N_2 in the Haber process
- When only nitrogen and hydrogen are present at the beginning of the reaction, the rate of the forward reaction is at its **highest**, since the **concentrations** of hydrogen and nitrogen are at their **highest**

- As the reaction proceeds, the concentrations of hydrogen and nitrogen gradually decrease, so the rate of the forward reaction will **decrease**
- However, the concentration of ammonia is gradually increasing and so the rate of the **backward** reaction will increase (ammonia will decompose to reform hydrogen and nitrogen)
- Since the two reactions are interlinked and none of the gas can escape, the rate of the forward reaction and the rate of the backward reaction will eventually become **equal** and equilibrium is reached:



YOUR NOTES



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Diagram showing when the rates of forward and backward reactions become equal

- Equilibrium position refers to the relationship between the concentration of reactants and products at the equilibrium state
- When the position of equilibrium shifts to the **left**, it means the concentration of **reactant** increases
- When the position of equilibrium shifts to **right**, this means the concentration of **product** increases

Le Chatelier's Principle

YOUR NOTES



EXTENDED

- The relative amounts of all the reactants and products at equilibrium depend on the conditions of the reaction
- This balance is framed in an important concept known as Le Chatelier's Principle, named after Henri Le Chatelier who was a French military engineer in the 19th century
- This principle states that when a change is made to the conditions of a system at equilibrium, the system automatically moves to **oppose** the change
- The principle is used to predict changes to the position of equilibrium when there are changes in **temperature, pressure** or **concentration**
- Knowing the energy changes, states and concentrations involved allows us to use the principle to manipulate the outcome of reversible reactions
- For example, if the pressure is increased, the position of equilibrium moves in the direction which has the smallest amount of gaseous molecules
- The position of equilibrium is said to shift to the **right** when the **forward** reaction is favoured and there is an increase in the amount of **products** formed
- The position of equilibrium is said to shift to the **left** when the **reverse** reaction is favoured and there is an increase in the amount of **reactants** formed

The Effect of Temperature on Equilibrium

- When the equilibrium mixture is heated, it becomes **dark brown** in colour. You can use this observation to deduce whether the backward reaction is exothermic or endothermic
- Equilibrium has shifted to the left as the colour dark brown means that more ICl is produced
- Increasing temperature moves the equilibrium in the endothermic direction
- **So the backward reaction is endothermic**

Table showing the Effects of Temperature on Equilibrium

CHANGE	HOW THE EQUILIBRIUM SHIFTS
INCREASE IN TEMPERATURE	EQUILIBRIUM MOVES IN THE ENDOTHERMIC DIRECTION TO REVERSE THE CHANGE
DECREASE IN TEMPERATURE	EQUILIBRIUM MOVES IN THE EXOTHERMIC DIRECTION TO REVERSE THE CHANGE

The Effect of Pressure on Equilibrium

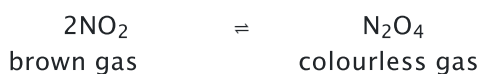
Table showing the Effects of Pressure on Equilibrium

YOUR NOTES



CHANGE	HOW THE EQUILIBRIUM SHIFTS
INCREASE IN PRESSURE	EQUILIBRIUM SHIFTS IN THE DIRECTION THAT PRODUCES THE SMALLER NUMBER OF MOLECULES OF GAS TO DECREASE THE PRESSURE AGAIN
DECREASE IN PRESSURE	EQUILIBRIUM SHIFTS IN THE DIRECTION THAT PRODUCES THE LARGER NUMBER OF MOLECULES OF GAS TO INCREASE THE PRESSURE AGAIN

Example: Nitrogen dioxide can form dinitrogen tetroxide, a colourless gas



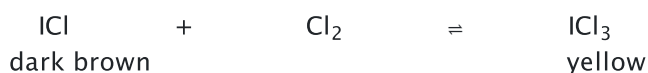
- Predict the effect of an increase in pressure on the position of equilibrium:
 - Number of molecules of gas on the left = 2
 - Number of molecules of gas on the right = 1
- An **increase** in pressure will cause equilibrium to shift in the direction that produces the **smaller** number of molecules of gas
- So equilibrium shifts to the right**
- The reaction mixture becomes paler as more colourless N_2O_4 is produced

The Effect of Concentration on Equilibrium

Table showing the Effects of Concentration on Equilibrium

CHANGE	HOW THE EQUILIBRIUM SHIFTS
INCREASE IN CONCENTRATION	EQUILIBRIUM SHIFTS TO THE RIGHT TO REDUCE THE EFFECT OF INCREASE IN THE CONCENTRATION OF A REACTANT
DECREASE IN CONCENTRATION	EQUILIBRIUM SHIFTS TO THE LEFT TO REDUCE THE EFFECT OF A DECREASE IN REACTANT (OR AN INCREASE IN THE CONCENTRATION OF PRODUCT)

Example: Iodine monochloride reacts reversibly with chlorine to form iodine trichloride



- Predict the effect of an increase in concentration on the position of equilibrium:
 - An increase in the concentration of ICl or Cl_2 causes the equilibrium to shift to the **right** so more of the **yellow** product is formed

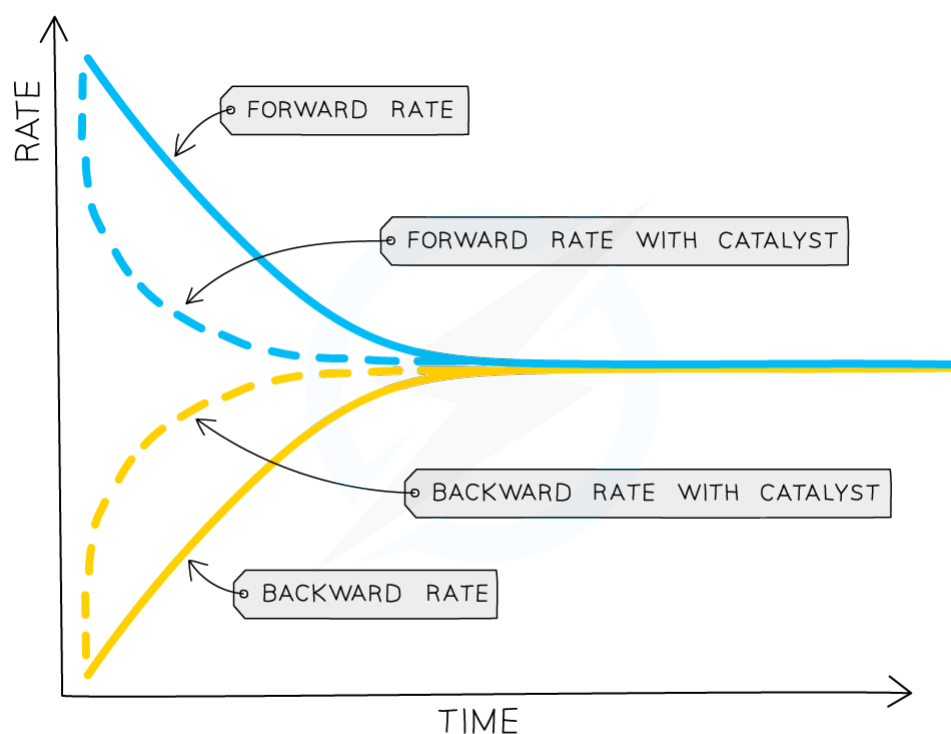
- A decrease in the concentration of ICl or Cl_2 causes the equilibrium to shift to the **left** so more of the **dark brown** reactant is formed

YOUR NOTES



The Effect of a Catalyst on Equilibrium

- The presence of a catalyst does **not** affect the position of equilibrium but it does increase the rate at which equilibrium is reached
- This is because the catalyst increases the rate of **both** the forward and backward reactions by the same amount (by providing an alternative pathway requiring lower activation energy)
- As a result, the **concentration** of reactants and products is nevertheless the **same** at equilibrium as it would be without the catalyst



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Diagram showing the effect of a catalyst on the time taken for equilibrium to be established



Exam Tip

When the conditions at equilibrium are changed, the system always responds by doing the **opposite**. For example if the concentration is increased the system tries to reduce it by changing the direction of the reaction or if the temperature is increased the system will try to reduce the temperature by absorbing the extra heat.

6.2.3 The Haber Process

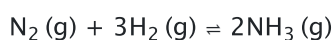
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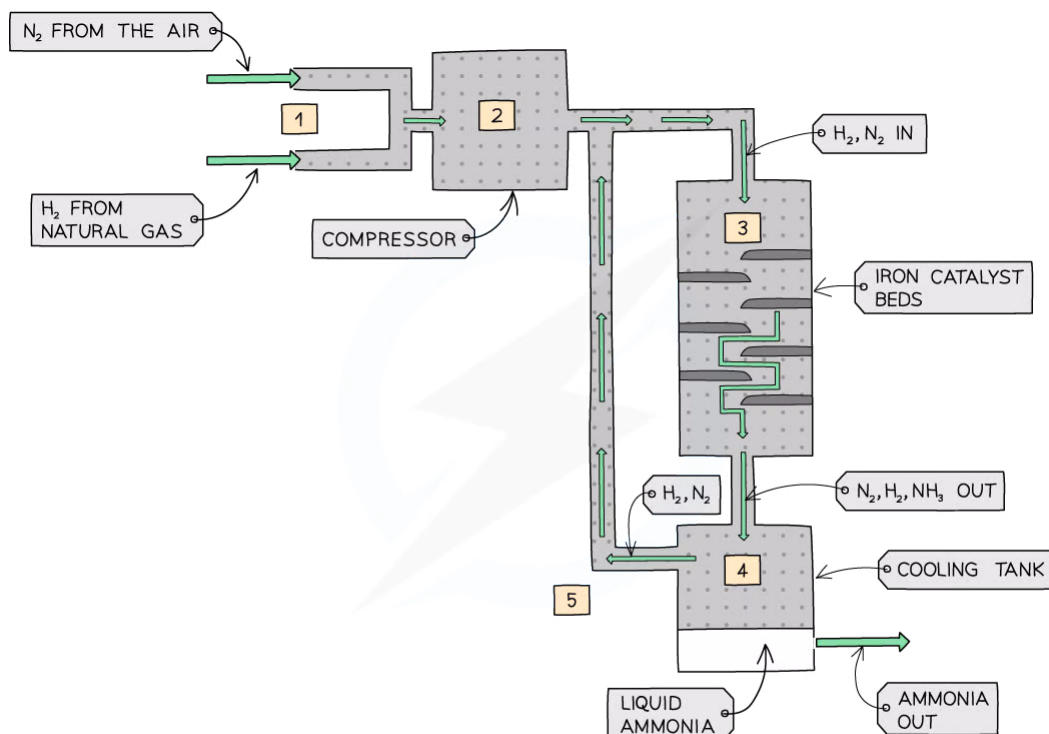
The Haber Process

EXTENDED

- Ammonia is manufactured using **The Haber process** which occurs in five stages:
- Stage 1:** H_2 and N_2 are obtained from natural gas and the air respectively and are pumped into the compressor through pipe
- Stage 2:** The gases are compressed to about 200 atmospheres inside the compressor
- Stage 3:** The pressurised gases are pumped into a tank containing layers of catalytic iron beds at a temperature of 450°C . Some of the hydrogen and nitrogen react to form ammonia:



- Stage 4:** Unreacted H_2 and N_2 and the product ammonia pass into a cooling tank. The ammonia is liquefied and removed to pressurised storage vessels
- Stage 5:** The unreacted H_2 and N_2 gases are recycled back into the system and start over again



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Exam Tip

You need to be able to recall the temperature, pressure and name of the catalyst used in the Haber process.

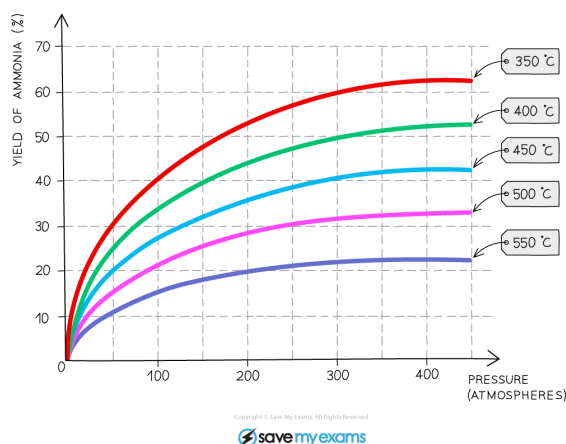
YOUR NOTES



Explaining the Conditions in the Haber Process

Extended

- Reaction conditions such as temperature and pressure affect the **rate** of a reaction
- If the reaction is reversible then the **position of equilibrium** is also affected by changes in these conditions and often we must consider a trade-off between the rate of reaction and product yield
- The graph below illustrates the effects of changing temperature and pressure on the yield of ammonia obtained
- By following any of the curved lines on the graph it can be seen that as the **pressure increases**, so too does the **yield** at any given temperature
- By following any vertical line upwards from the x-axis, the graph shows that as the temperature **decreases**, the yield actually **increases**
- The actual conditions used must be chosen depending on a number of economical, chemical and practical considerations



The yield of ammonia produced changes with changes made to temperature and pressure

Economic Considerations

- Like all industries, companies that manufacture and sell chemical goods do so to make a profit
- Part of the industrial process is the economic decision on how and where to design and implement a manufacturing site
- The availability and cost of **raw materials** is a major consideration which must be studied well before any decisions are taken
- In the Haber Process the raw materials are readily available and inexpensive to purify:
 - **Nitrogen** – from the **air**
 - **Hydrogen**– from **natural gas**
- If the cost of extraction of raw materials is too high or they are unavailable then the process is no longer **economically viable**
- Many industrial processes require huge amounts of **heat** and **pressure** which is very expensive to maintain

YOUR NOTES



- Production energy costs are also a factor to be considered carefully and alongside the raw materials issue

YOUR NOTES



Temperature: 450 °C

- A **higher** temperature would favour the reverse reaction as it is endothermic (takes in heat) so a higher yield of **reactants** would be made
- If a **lower** temperature is used it favours the forward reaction as it is exothermic (releases heat) so a higher yield of **products** will be made
- However at a lower temperature the rate of reaction is very **slow**
- So 450 °C is a compromise temperature between having a **lower yield** of products but being made more quickly

Pressure: 200 atm

- A **lower** pressure would favour the reverse reaction as the system will try to increase the pressure by creating more molecules (4 molecules of gaseous reactants) so a higher yield of **reactants** will be made
- A **higher** pressure would favour the forward reaction as it will try to decrease the pressure by creating less molecules (2 molecules of gaseous products) so a higher yield of **products** will be made
- However high pressures can be dangerous and very expensive equipment is needed
- So 200 atm is a **compromise** pressure between a lower yield of products being made **safely** and **economically**

Catalyst: Iron

- The presence of a catalyst does **not** affect the position of equilibrium but it does increase the **rate** at which equilibrium is reached
- This is because the catalyst increases the rate of **both** the forward and backward reactions by the same amount (by providing an alternative pathway requiring **lower activation energy**)
- As a result, the **concentration** of reactants and products is nevertheless the **same** at equilibrium as it would be without the catalyst.
So a catalyst is used as it helps the reaction reach equilibrium quicker
- It allows for an acceptable yield to be achieved at a **lower temperature** by lowering the activation energy required
- Without it the process would have to be carried out at an even **higher temperature**, increasing **costs** and decreasing **yield** as the higher temperature **decomposes** more of the NH_3 molecules



Exam Tip

The reaction conditions chosen for the Haber process are not ideal in terms of the yield but do provide balance between product yield, reaction rate and production cost. These are called **compromise conditions** as they are chosen to give a good compromise between the yield, rate and cost.

6.2.4 The Contact Process

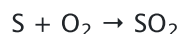
YOUR NOTES



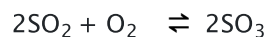
The Contact Process

EXTENDED

- Sulfuric acid is synthesised by the **Contact process**
- Concentrated sulfuric acid is used in car batteries, making **fertilisers, soaps and detergents**
- The first stage is the production of sulfur dioxide, either by burning sulfur to oxidise the sulfur (equation shown below), or roasting sulfide ores



- The **main stage** in the Contact process is the oxidation of sulfur dioxide to sulfur trioxide using a **vanadium(V) oxide, V_2O_5 , catalyst**:



- The oxygen used in this stage is obtained from air
- The conditions for this main stage of production are:
 - A temperature of **450 °C**
 - A pressure of **2 atm (200 kPa)**
- Once sulfur trioxide is formed, it undergoes more processes to produce sulfuric acid



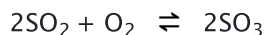
Exam Tip

You need to recall the temperature, pressure and catalyst needed for the Contact process and the equation for the main stage only.



Explaining the Conditions in the Contact Process

- Similar to the Haber process, the pressure and temperature used need to be considered
- The equation for the main stage of the Contact process is:



Temperature: 450°C

- The forward reaction is **exothermic**, so increasing the temperature shifts the position of equilibrium to the left in the direction of the **reactants**
- Therefore the higher the temperature, the **lower** the yield of sulfur trioxide
- The optimum temperature is a compromise between a higher rate of reaction at a higher temperature and a lower equilibrium yield at a higher temperature

Pressure: 2 atm

- An increase in pressure shifts the position of equilibrium to the **right** in the direction of a smaller number of gaseous molecules
- However the position of equilibrium lies far to the right (the equilibrium mixture contains about 96% sulfur trioxide)
- So the reaction is carried out at just above atmospheric pressure because:
 - High pressures can be dangerous and very expensive equipment is needed
 - A higher pressure causes the sulfur dioxide to liquefy



Exam Tip

Remember: These conditions are a **compromise** between yield, rate, safety and cost.

6.3 Redox

6.3.1 Oxidation & Reduction

YOUR NOTES



Oxidation & Reduction

Redox reactions

- Oxidation and reduction take **place together** at the **same time** in the **same reaction**
- These are called **redox** reactions
- **Oxidation** is a reaction in which **oxygen is added** to an element or a compound
- **Reduction** is a reaction in which **oxygen is removed** from an element or compound

Example: Identifying the loss and gain of oxygen in an equation

zinc oxide + carbon $\rightarrow$ zinc + carbon monoxide



- In this reaction, the zinc oxide has been reduced since it has **lost** oxygen
- The carbon atom has been oxidised since it has **gained** oxygen

Names using oxidation numbers

- Transition elements can bond in different ways by forming ions with different charges
- When naming, the charge on the ion is shown by using a Roman numeral after the element's name
 - e.g. iron can form ions with a 2+ charge, called **iron(II)** ions or a 3+ charge, called **iron(III)** ions
- The Roman numeral is the **oxidation number** of the element
- When iron reacts with oxygen to form iron oxide, the formula depends on the oxidation state of the iron ions
 - The compound where iron has a 2+ charge has the formula FeO and is called iron(II) oxide
 - The compound where iron has a 3+ charge has the formula Fe_2O_3 and is called iron(III) oxide



Exam Tip

You may see the term oxidation state used instead of oxidation number. Although there is a subtle difference between the two terms (this is beyond the scope of this course), they are often used interchangeably. Usually oxidation number is used to refer to the Roman numerals found within the name.

6.3.2 Redox & Electron Transfer

YOUR NOTES



Redox & Electron Transfer

EXTENDED

Oxidation & Reduction

- Redox reactions can also be defined in terms of electron transfer
- Oxidation** is a reaction in which an element, ion or compound **loses electrons**
 - The **oxidation number** of the element is **increased**
 - This can be shown in a **half equation**, e.g. when silver reacts with chlorine, silver is **oxidised** to silver ions:



- Reduction** is a reaction in which an element, ion or compound **gains electrons**
 - The **oxidation number** of the element is **decreased**
 - This can be shown in a half equation, e.g. when oxygen reacts with magnesium, oxygen is **reduced** to oxide ions:

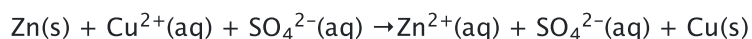


Example: Identifying Redox Reactions

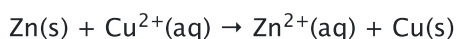
zinc + copper sulphate → zinc sulphate + copper



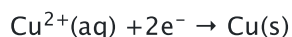
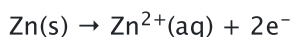
- The ions present (with state symbols) in the equation are:



- The **spectator ions** (those that do not change) are $\text{SO}_4^{2-}(\text{aq})$
- These can be removed and the ionic equation written as:



- By analysing the ionic equation, we can split the reaction into two half equations by adding in the electrons to show how the changes in charge have occurred:



- It then becomes clear that zinc has been oxidised as it has **lost electrons**
- Copper ions have been reduced as they have **gained electrons**



Exam Tip

Use the mnemonic **OIL-RIG** to remember oxidation and reduction in terms of the movement of electrons: **O**xidation **I**s **L**oss – **R**eduction **I**s **G**ain.

Identifying Redox Reactions

YOUR NOTES



EXTENDED

Oxidation Number

- The **oxidation number** (also called oxidation state) is a number assigned to an atom or ion in a compound which indicates the **degree of oxidation** (or reduction)
- It shows the number of electrons that an atom has lost, gained or shared in forming a compound
- The oxidation number helps you to keep track of the movement of electrons in a redox process
- It is written as a +/– sign followed by a number (not to be confused with charge which is written by a number followed by a +/– sign)
- E.g. aluminium in a compound usually has the oxidation state +3
- A few simple rules help guide you through the process of determining the oxidation number of any element

Table of Rules for Assigning Oxidation Numbers

YOUR NOTES



Rule	Example
1. The oxidation number of any uncombined element is zero	H_2 Zn O_2
2. Many atoms or ions have fixed oxidation number in compounds	Group 1 elements are always +1 Group 2 elements are always +2 Fluorine is always -1 Hydrogen is +1 (except for in metal hydrides like NaH, where it is -1) Oxygen is -2 (except in peroxides, where it is -1 and F_2O where it is +2)
3. The oxidation number of an element in mono-atomic ion is always same as the charge	Zn^{2+} oxidation number = +2 Fe^{3+} oxidation number = +3 Cl^- oxidation number = -1
4. The sum of the oxidation number in a compound is zero	NaCl oxidation number of Na = +1 oxidation number of Cl = -1 sum oxidation number = 0
5. The sum of oxidation number in ion is equal to the charge on the ion	SO_4^{2-} oxidation number of S = +6 oxidation number of 4 O atoms = $4 \times (-2)$ sum oxidation number = -2

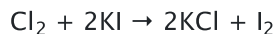
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- Redox reactions can be identified by the changes in the **oxidation number** when a reactant goes to a product



Worked Example

The equation for the reaction between chlorine and potassium iodide is shown below.



Identify which species has been:

- a) Oxidised
- b) Reduced

Answer:

- The species that has been oxidised is iodine
 - The oxidation number of I^- is -1
 - The oxidation number of iodine in I_2 is 0
 - The oxidation number has increased so the iodine has been oxidised (lost electrons)
 - $2\text{I}^-(\text{aq}) \rightarrow \text{I}_2(\text{s}) + 2\text{e}^-$
- The species that has been reduced is chloride ions
 - The oxidation number of chlorine as Cl_2 is 0 .
 - The oxidation number of Cl^- is -1
 - The oxidation number has decreased so the Cl^- has been reduced (gained electrons)
 - $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$

Identifying Redox Reactions by Colour Changes

- The tests for redox reactions involve the observation of a colour change in the solution being analysed
- Two common examples are acidified potassium manganate(VII), and potassium iodide
- Potassium manganate(VII), KMnO_4 , is an **oxidising agent** which is often used to test for the presence of **reducing agents**
- When acidified potassium manganate(VII) is added to a reducing agent its colour changes from **purple** to **colourless**

YOUR NOTES

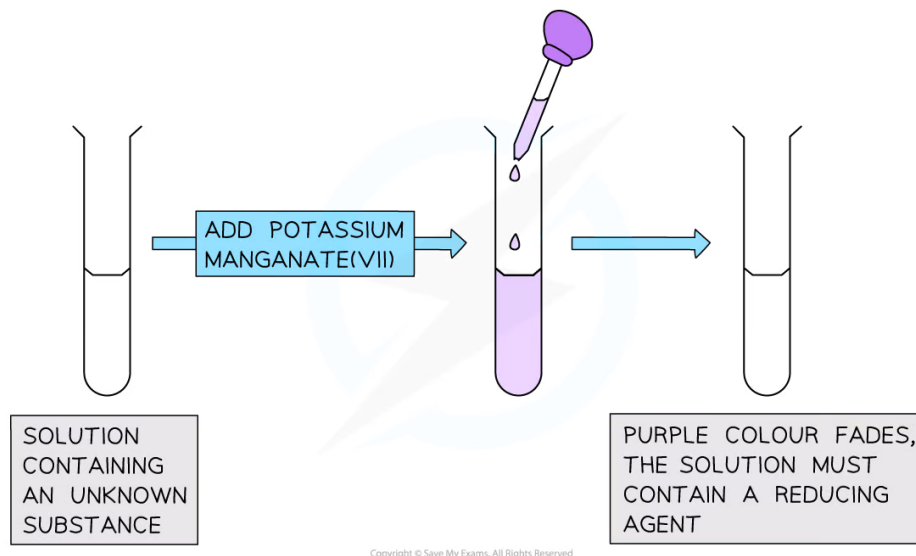
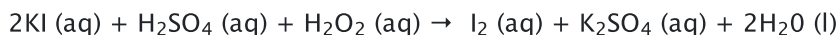


Diagram to show the colour change when potassium manganate(VII) is added to a reducing agent

- Potassium iodide, KI, is a **reducing** agent which is often used to test for the presence of **oxidising** agents
- When added to an acidified solution of an oxidising agent such as aqueous chlorine or hydrogen peroxide (H_2O_2), the solution turns a **red-brown** colour due to the formation of iodine, I_2 :



- The potassium iodide is oxidised as it loses electrons and hydrogen peroxide is reduced, therefore potassium iodide is acting as a **reducing agent** as it will itself be oxidised:

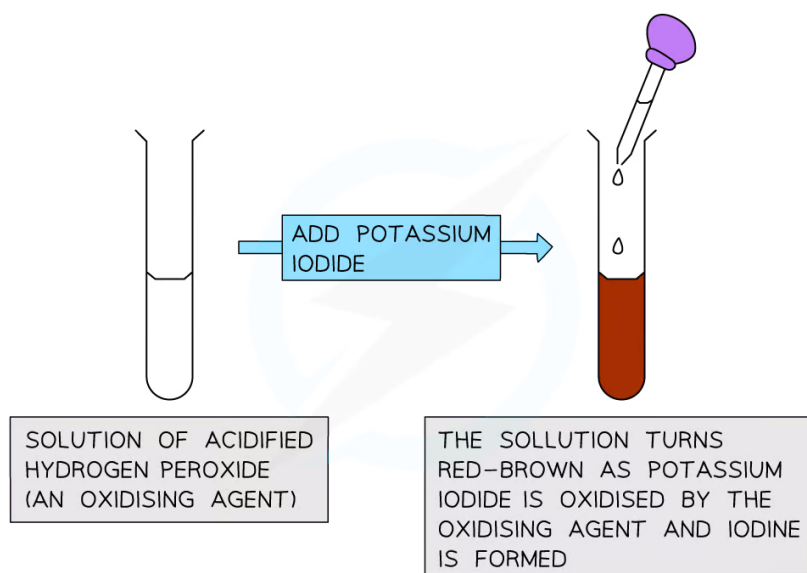
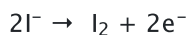


Diagram to show the colour change when potassium iodide is added to an oxidising agent

YOUR NOTES



Oxidising & Reducing Agents

YOUR NOTES



EXTENDED

Oxidising agent

- A substance that **oxidises** another substance, and becomes **reduced** in the process
- An oxidising agent **gains** electrons as another substance loses electrons
- Common examples include hydrogen peroxide, fluorine and chlorine

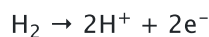
Reducing agent

- A substance that **reduces** another substance, and becomes **oxidised** in the process
- A reducing agent **loses** electrons as another substance gains electrons
- Common examples include carbon and hydrogen
- The process of reduction is very important in the chemical industry as a means of extracting metals from their ores

Example



- In the above reaction, hydrogen is reducing the CuO and is itself oxidised as it has lost electrons, so the **reducing agent** is therefore **hydrogen**:

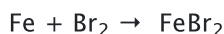


- The CuO is reduced to Cu by gaining electrons and has oxidised the hydrogen, so the **oxidising agent** is therefore **copper oxide**



Worked Example

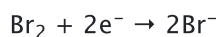
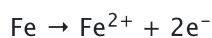
When iron reacts with bromine to form iron(II) bromide, a redox reaction occurs:



What is acting as the reducing agent in this reaction?

Answer

Step 1 – Write half equations to work out what has gained/lost electrons



Fe loses electrons; Br₂ gains electrons

Step 2 – Deduce what has been oxidised/reduced (remember OIL RIG)

Fe has been oxidised as it has lost electrons

Br₂ has been reduced as it has gained electrons

Step 3 – Identify the reducing agent

Fe is the reducing agent as it has been oxidised by losing electrons and caused Br_2 to be reduced as it gained electrons

YOUR NOTES



